

PAPER 1 / QUESTION 4

8820-6501 / original question 4

4. An object of mass 1 kg is thrown downwards from a height of 20 m. The initial speed of the object is 6 m s^{-1} . The object hits the ground at a speed of 20 m s^{-1} . Assume $g = 10 \text{ m s}^{-2}$. What is the best estimate of the energy transferred from the object to the air as it falls?
- A. 6 J
 - B. 18 J
 - C. 182 J
 - D. 200 J